CHAPTER 1

Metabolic response to injury

Learning objectives

To understand:

- How the body responds to accidental injury and surgery
- Physiological and biochemical changes that occur during injury and recovery
- Mediators and pathways of the metabolic response to injury
- Avoidable factors that compound the metabolic response to injury
- How the metabolic response to injury influences surgical outcomes
- Concepts behind optimal perioperative care

INTRODUCTION

As surgeons we are inextricably linked with tissue injury and its effects, both from the damage which operating inevitably causes and from the treatment of accidental traumatic injury. The body responds to significant local tissue injury, whether surgical or accidental, with a series of systemic changes which affect the functions of vital organs. This surgical stress response is brought about by several pathways involving hormones, inflammation-related cytokines and neural circuits. It leads to alterations in body metabolism, wound healing and immunity and in the function of specific organs. These changes are known collectively as the metabolic response to injury. While these responses are designed to limit damage and begin repair processes, not all the effects are beneficial by any means. They can lead to complications, especially sepsis, which can then amplify and prolong the abnormal processes and lead to or prolong multiple organ dysfunction syndrome (MODS). Given that these metabolic effects of injury can have a significant impact on recovery and survival from many types of surgery and surgical illness, surgeons require an understanding of them in order to care optimally for their patients. Successful management of the metabolic response improves outcomes and forms the basis of modern perioperative care after major surgery as well as the treatment of severely injured and septic patients. This chapter will look primarily at the metabolic responses to injury while shock, fluid balance, sepsis and nutrition are covered in greater depth in *Chapters 2 and 25*.

Homeostasis

Homeostasis is the concept of maintaining a constant internal environment that allows cellular processes to function optimally. Many aspects of surgery, trauma and injury affect homeostasis and can lead to organ dysfunction. Traditionally the metabolic response to injury is divided into an initial period of catabolism (which may include a period of shock) followed by an anabolic phase of repair and tissue healing.

The catabolic phase begins at the time of injury and is characterised by hypovolaemia, decreased basal metabolic rate, reduced cardiac output, hypothermia and lactic acidosis. The main physiological role of this phase is to conserve both circulating volume and energy stores and thus maximise survival chances for future recovery. A series of neurohormonal responses accompany these effects and trigger a systemic inflammatory response syndrome (SIRS), where body stores are mobilised for recovery and repair. The catabolic effects include muscle breakdown, weight loss and hyperglycaemia, which themselves increase the risk of complications, especially sepsis. As the catabolic phase subsides, an anabolic (rebuilding) phase develops, which may last for weeks if extensive recovery and repair are required following serious injury.

Modern surgical care

The role of surgical critical care, including resuscitation and/or organ support, must be to work alongside the metabolic effects of injury while the patient is restored to a situation from which homeostatic mechanisms can achieve a return to normality. The systemic effects of injury still impact heavily on survival and complications through loss of muscle mass, sepsis and MODS. In fact, modern treatment of major trauma can now be so successful that the great majority of hospital deaths in developed countries occur after some days as a result of complex physiological processes, rather than as a direct and rapid consequence of organ damage or blood loss, although it is the initial injury and blood loss that sets the scene for the later systemic effects. Parallel with the catabolic effects introduced above, inflammatory-type processes cause immune suppression. While this inflammation is often initially sterile, the nature of surgery and injury predisposes to infection and sepsis. Impaired immunity as part of the metabolic response

compounds this risk and explains why sepsis and MODS/failure is a key part of perioperative care and a leading mode of death among our patients. Even in modern trauma systems, MODS carries a mortality of around 25%.

As a consequence of modern understanding of the metabolic response to injury, elective surgical practice now seeks to actively reduce the need for a homeostatic response by minimising the primary insult via minimal access surgery and by 'stress-free' perioperative care or enhanced recovery after surgery (ERAS). This chapter will review the mediators of the stress response, the physiological and biochemical pathway changes associated with surgical injury and the changes in body composition that occur following surgical injury. Emphasis is placed on why knowledge of these events is important to understand the rationale for modern 'stress-free' perioperative and critical care.

Summary box 1.1

Basic concepts

- Homeostasis is the foundation of normal physiology
- 'Stress-free' perioperative care helps to preserve homeostasis following elective surgery
- Resuscitation, surgical intervention and critical care can return the severely injured patient to a situation in which homeostasis becomes possible once again
- The metabolic response to surgery influences these processes profoundly, particularly through catabolic effects, MODS and impaired immunity

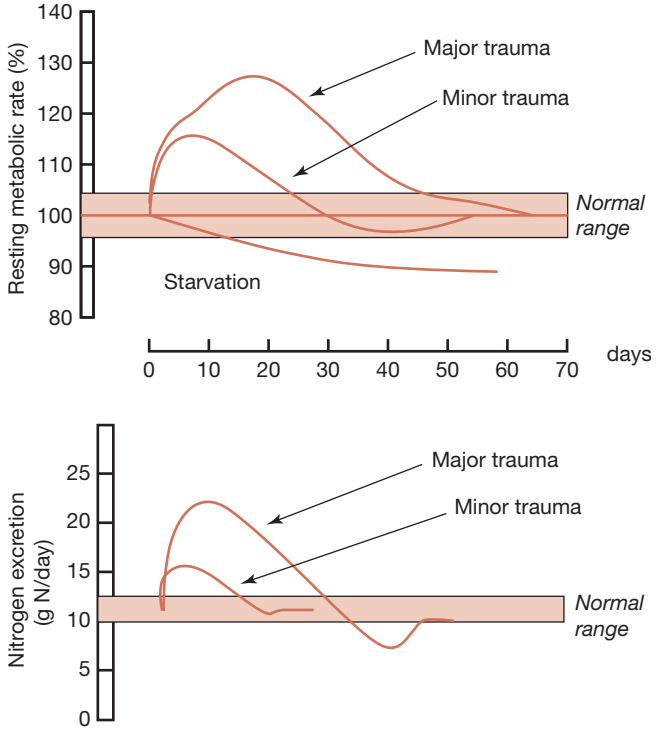

Figure 1.1 Hypermetabolism and increased nitrogen excretion are closely related to the magnitude of the initial injury and show a graded response.

THE MAGNITUDE OF THE INJURY RESPONSE

It is important to recognise that, in general or population terms, the metabolic response to injury is graded: the more severe the injury, the greater the response (*Figure 1.1*). This concept applies not only to physiological and metabolic changes but also to immunological changes and other sequelae. Thus, following major elective surgery, there may be a transient and modest rise in temperature, heart rate, respiratory rate, energy expenditure and peripheral white cell count. Following major trauma, emergency surgery, sepsis or burns, these changes are accentuated, resulting in SIRS, with hypermetabolism, marked catabolism, shock and even MODS. However, genetic variability also plays a key role in determining the intensity of the inflammatory response, with some individual patients responding much more dramatically than others to apparently similar conditions.

MEDIATORS OF THE METABOLIC RESPONSE TO INJURY

Tissue damage and inflammation

Tissue injury is sensed in several ways. Tissue damage causes the release of cellular and other molecular fragments known as damage-associated molecular patterns (DAMPs) or alarmins. These DAMPs are sensed by pattern recognition receptors (PRRs), such as Toll-like receptors and NOD-like receptors (or nucleotide-binding leucine-rich repeat receptors) on cells of the innate immune system, which includes macrophages, neutrophils and dendritic cells. These cells are attracted and activated, triggering the formation of complex intracellular proteins known as inflammasomes. This results in the activation of caspases; these are enzymes that, in turn, activate key inflammatory cytokines including interleukin-1 (IL-1), IL-6 and many others. PRR activation also leads to release of tumour necrosis factor alpha (TNF), interferons, chemokines and other mediators. Thus begins a sterile systemic inflammatory cascade that leads to local inflammation and, when sufficiently severe, to a clinically detectable SIRS. Once activated by DAMPs, inflammasomes also contribute to cell death, tissue damage and immune suppression. DAMPs can activate inflammasome formation in endothelial cells and platelets, resulting in leaky capillaries and coagulopathy; these are changes that can result in the production of more DAMPs owing to local ischaemia from microcirculatory effects. Local inflammation begins the process of tissue repair but SIRS, when uncontrolled or prolonged, becomes a risk factor for acute kidney injury, acute lung injury and coagulopathy, and hence for MODS and organ failure. Within the injured brain, secondary brain injury can occur.

DAMPs thought to be important in tissue trauma include heat shock proteins, high mobility group protein B1 (HMGB1), S100 proteins and fragments of nucleic acids. Commonly, DAMPs can activate several different receptors and pathways. This crossover, or redundancy as it is termed, is a characteristic of inflammation and has been one of the barriers to developing

effective therapeutic blockade of these mechanisms. Furthermore, DAMPs can be self-perpetuated during the complicated course of a surgical critical illness, amplifying and prolonging the inflammatory process and related organ dysfunction. Triggers to further release of DAMPs include sepsis, haemorrhage, massive transfusion, acidosis, surgery, crush syndrome and ischaemia–reperfusion. Thus the secondary insults of delayed or ineffective treatment of complications such as ongoing bleeding, ischaemia or sepsis will tend to maintain and amplify the inflammatory process and its resulting immune dysfunction. This can become a prolonged or self-perpetuating process (*Table 1.1*).

Summary box 1.2

Neuroendocrine response to injury/critical illness

The neuroendocrine response to severe injury/critical illness is biphasic:

- **Acute phase** (hours) characterised by elevated counter-regulatory hormones (cortisol, glucagon, adrenaline). Changes are thought to be beneficial for short-term survival
- **Chronic phase** (days) associated with hypothalamic suppression and low serum levels of the respective target organ hormones. Changes may contribute to chronic wasting

TABLE 1.1 Some secondary triggers of the metabolic response to injury.

Secondary triggers of inflammatory pathways in trauma and surgery
• Sepsis
• Haemorrhage
• Massive transfusion
• Acidosis
• Surgery
• Crush syndrome
• Ischaemia–reperfusion
These events can amplify or prolong the catabolic phase, leading to organ failure or immune dysfunction.

Neuroendocrine response to injury

Patients also respond rapidly to injury by the classical neuroendocrine pathways of the stress response, consisting of afferent nociceptive neurones, the spinal cord, thalamus, hypothalamus and pituitary (*Figure 1.2*). Nociceptive neurones are excited by the effects of local inflammation as well as by direct injury. The neurones terminate in the hypothalamus and release corticotropin-releasing factor (CRF). CRF stimulates adrenocorticotropic hormone (ACTH) release from the anterior pituitary, which then acts on the adrenals to increase the secretion of cortisol within hours of injury. Hypothalamic activation of the sympathetic nervous system causes release of adrenaline (epinephrine) and also stimulates release of glucagon. An intravenous infusion of a cocktail of these 'counter-regulatory' hormones (glucagon, glucocorticoids and catecholamines) reproduces many aspects of the metabolic response to injury. The metabolic effects of the acute rise in the levels of these hormones is to liberate glucose from carbohydrate stores and to begin the breakdown of fat and protein as metabolic substrates for energy and repair. There are, however, many other effects, including alterations in insulin release and sensitivity, hypersecretion of prolactin and growth hormone (GH) in the presence of low circulatory insulin-like growth factor-1 (IGF-1) and inactivation of peripheral thyroid hormones and gonadal function. Of note, GH has direct lipolytic, insulin-antagonising and proinflammatory properties.

As described above, the innate immune system (principally macrophages), once activated by DAMPs, interacts in a complex manner with the adaptive immune system (T cells, B cells) in co-generating the metabolic response to injury (*Figure 1.2*). Proinflammatory cytokines including IL-1, TNF alpha (TNFα), IL-6 and IL-8 are produced within the first 24 hours and act directly on the hypothalamus to cause pyrexia. Such cytokines also augment the hypothalamic stress response and act directly on skeletal muscle to induce proteolysis while inducing acute-phase protein production in the liver. Proinflammatory cytokines also play a complex role in the development of peripheral insulin resistance. Other important proinflammatory mediators include nitric oxide ([NO] via inducible nitric oxide synthetase [iNOS]) and a variety of prostanoids (via cyclooxygenase-2 [Cox-2]). Changes in organ function (e.g. renal hypoperfusion/impairment) may be induced by excessive vasoconstriction via endogenous factors such as endothelin-1. Complement and kinin pathways are also activated and processes of programmed cell death and phagocytosis are triggered to clear damaged tissues.

There are many complex interactions among the neuroendocrine, cytokine and metabolic axes. For example, although cortisol is immunosuppressive at high levels, it acts synergistically with IL-6 to promote the hepatic acute-phase response. ACTH release is enhanced by proinflammatory cytokines and the noradrenergic system. The resulting rise in cortisol levels may form a weak feedback loop, attempting to limit the proinflammatory stress response. Finally, hyperglycaemia may aggravate the inflammatory response in the mitochondria, causing the formation of excess oxygen free radicals and also altering gene expression to enhance cytokine production.

At the molecular level, the changes that accompany systemic inflammation are extremely complex. In one study using network-based analysis of changes in mRNA expression in leukocytes following exposure to endotoxin, there were changes in the expression of more than 3700 genes, with over half showing decreased expression and the remainder increased expression. The cell surface receptors, signalling mechanisms and transcription factors that initiate these events are also complex. Although the detailed mechanisms are being steadily identified, specific molecular therapies remain elusive and certainly subservient to optimal clinical care.

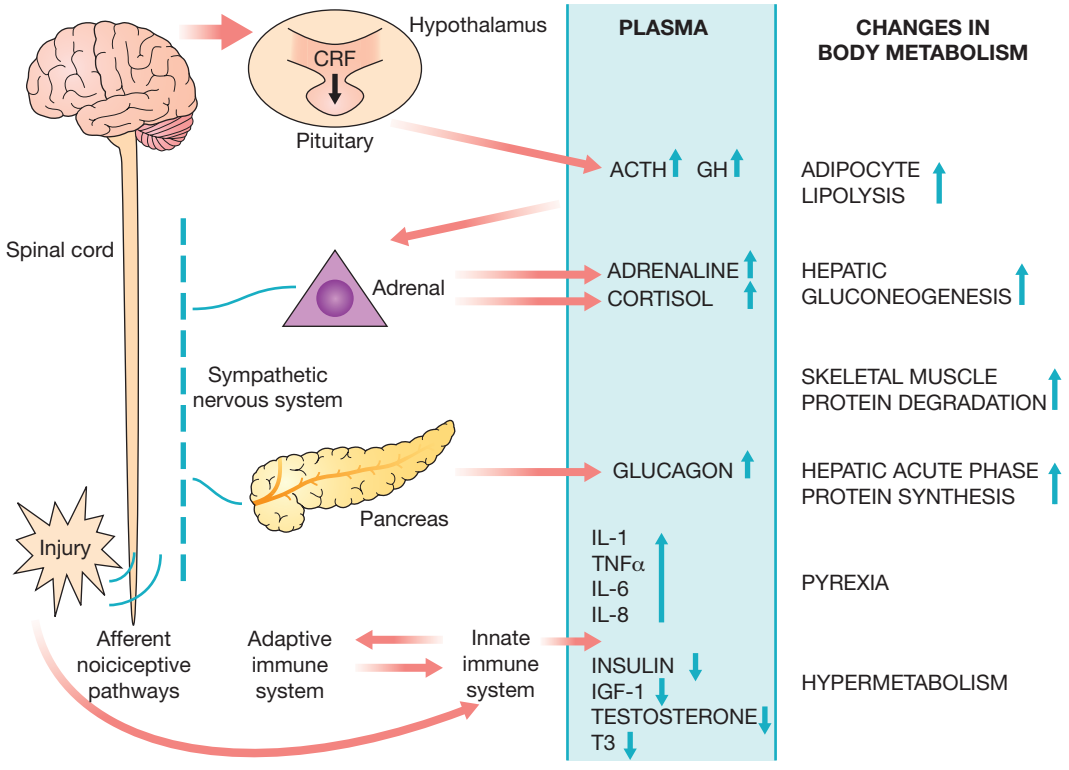

Figure 1.2 The integrated response to surgical injury (first 24–48 hours): there is a complex interplay between the neuroendocrine stress response and the proinflammatory cytokine response of the innate immune system. ACTH, adrenocorticotropic hormone; GH, growth hormone; IGF, insulin-like growth factor; IL, interleukin; T3, triiodothyronine; TNFα, tumour necrosis factor alpha.

Agonists and antagonists: an uncertain balance

Within hours of the upregulation of proinflammatory cytokines, endogenous cytokine antagonists enter the circulation (e.g. interleukin-1 receptor antagonist [IL-1Ra] and TNF-soluble receptors [TNF-sR-55 and 75]) and act to control the initial proinflammatory response and limit any systemic organ damage caused by it. A complex further series of adaptive changes includes the development of a counter-inflammatory response regulated by IL-4, -5, -9 and -13 and transforming growth factor beta (TGFβ). Within inflamed tissue the duration and magnitude of acute inflammation as well as the return to homeostasis are influenced by a group of local mediators known as specialised pro-resolving mediators (SPMs), which include essential fatty acid-derived lipoxins, resolvins, protectins and maresins. These endogenous resolution agonists orchestrate the uptake and clearance of apoptotic polymorphonuclear neutrophils and microbial particles, reduce proinflammatory cytokines and lipid mediators as well as enhance the removal of cellular debris. Thus, both at the systemic level (endogenous cytokine antagonists – see earlier) and at the local tissue level, the body attempts to limit the inflammatory response, but further tissue damage, sepsis or other complications challenge these processes of resolution. As with the initial inflammatory response to tissue injury, it appears that the degree of the secondary anti-inflammatory response varies between individuals, probably on a genetic basis. If the anti-inflammatory response dominates or is accentuated and prolonged in critical illness, it is characterised as a compensatory anti-inflammatory response syndrome (CARS), resulting in immunosuppression and an increased susceptibility to opportunistic (nosocomial) infection. Further sepsis, with its associated catabolism, results. CARS can be prolonged by ongoing critical illness as part of an ongoing vicious cycle of chronic critical illness (also known as Persistent Inflammation, Immunosuppression and Catabolism) syndrome. Thus both the initial inflammatory response to tissue injury and the secondary modulating responses can be seen to differing degrees in different individuals or at different stages of the critical illness. Either circumstance can cause harm, and rapid restoration of homeostasis and preventing secondary inflammation or sepsis are key therapeutic principles that influence late outcomes as well as immediate ones.

Summary box 1.3

The metabolic response to surgery and injury: key characteristics

- Rapid onset driven by proinflammatory cytokines (e.g. IL-1, IL-6 and TNFα)
- Broadly related to injury severity; most severe in sepsis, burns and major trauma
- Varies in severity between individuals (genetic)
- Causes catabolism, muscle breakdown, immunosuppression and organ dysfunction/failure
- Counterbalanced by antagonist response but the balance may be imperfect
- Prolonged by sepsis and other secondary insults
- Can become chronic
- Associated with most late deaths from injury or surgery in developed health systems

METABOLIC CHANGES AFTER SURGERY AND TRAUMA

The catabolic phase begins at the time of injury and lasts for approximately 24–48 hours. It may be attenuated by proper resuscitation and is characterised by hypovolaemia, decreased basal metabolic rate, reduced cardiac output, hypothermia and lactic acidosis. The predominant hormones regulating the catabolic phase are catecholamines, cortisol and aldosterone (following activation of the renin–angiotensin system). The magnitude of this neuroendocrine response depends on the degree of tissue damage, blood loss and the stimulation of somatic afferent nerves at the site of injury. The main physiological role of the catabolic phase is to conserve both circulating volume and energy stores for later recovery and repair.

Following resuscitation, the catabolic phase evolves into a hypermetabolic flow phase, which corresponds to SIRS. This phase involves the mobilisation of body energy stores for recovery and repair, and the subsequent replacement of lost or damaged tissue. It is characterised by tissue oedema (from vasodilatation and increased capillary leakage), increased basal metabolic rate (hypermetabolism), increased cardiac output, raised body temperature, leukocytosis, increased oxygen consumption and increased gluconeogenesis.

During the catabolic phase, the increased production of counter-regulatory hormones (including catecholamines, cortisol and glucagon) and inflammatory cytokines (e.g. IL-1, IL-6 and TNFα) results in significant fat and protein mobilisation, leading to significant weight loss and increased urinary nitrogen excretion. During shock, insulin levels do not rise as expected to combat the hyperglycaemia that occurs in response to stress hormone release and plasma insulin can even fall after severe injury. Within a few days, insulin production is increased but is associated with significant insulin resistance and, therefore, injured patients often exhibit poor glycaemic control. Importantly, the combination of pronounced or prolonged catabolism in association with insulin resistance places patients within this phase at increased risk of septic and other complications. Obviously, the development of complications will further aggravate the neuroendocrine and inflammatory stress responses, thus creating a vicious catabolic cycle and management of blood sugar levels remains an important step.

Summary box 1.4

Purpose of neuroendocrine changes following surgery or trauma

The constellation of neuroendocrine changes following surgery or trauma acts to:

- Provide essential substrates for survival from tissue breakdown
- Postpone anabolism
- Optimise host defence

These changes may be helpful in the short term, but may be harmful in the long term, especially to the severely injured or critically ill patient.

MANAGING THE CATABOLIC STRESS RESPONSE

There are several key elements that determine the extent of catabolism and thus govern the metabolic and nutritional care of the surgical patient. It must be remembered that, during the response to injury, not all tissues are catabolic. Indeed, the essence of this coordinated response is to allow the body to reprioritise limited resources away from peripheral tissues (muscle, adipose tissue, skin) and towards key viscera (liver, immune system) and the wound (*Figure 1.3*). However the damage to skeletal muscle can be catastrophic.

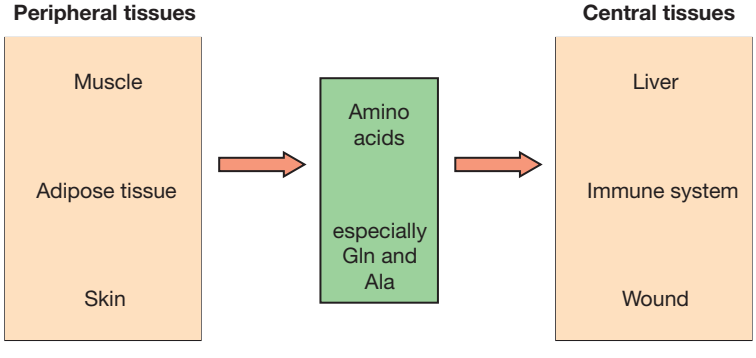

Figure 1.3 During the metabolic response to injury, the body reprioritises protein metabolism away from peripheral tissues and towards key central tissues such as the liver, immune system and wounds. One of the main reasons why the reutilisation of amino acids derived from muscle proteolysis leads to net catabolism is that the increased glutamine and alanine efflux from muscle is derived, in part, from the irreversible degradation of branched chain amino acids. Ala, alanine; Gln, glutamine.

Hypermetabolism

The majority of trauma patients (except possibly those with extensive burns, in whom a greater effect can be seen) demonstrate energy expenditures approximately 15–25% above predicted healthy resting values. The predominant cause appears to be a complex interaction between the central control of metabolic rate and peripheral energy utilisation. In particular, central thermodysregulation (caused by the proinflammatory cytokine cascade), increased sympathetic activity, abnormalities from wound circulation (ischaemic areas produce lactate, which must be metabolised by the adenosine triphosphate [ATP]-consuming hepatic Cori cycle; hyperaemic areas cause an increase in cardiac output), increased protein turnover and nutritional support may all increase patient energy expenditure. Theoretically, patient energy expenditure could rise even higher than observed levels following surgery or trauma, but several features of standard intensive care (including bed rest, paralysis, ventilation and external temperature regulation) limit the hypermetabolic driving forces of the stress response. Furthermore, the skeletal muscle wasting experienced by patients with prolonged catabolism actually limits the volume of metabolically active tissue (see ***Alterations in skeletal muscle protein metabolism***).

Alterations in skeletal muscle protein metabolism

Muscle protein is continually synthesised and broken down with a turnover rate in humans of 1–2% per day. Under normal circumstances, synthesis equals breakdown and muscle bulk remains constant. Physiological stimuli that promote net muscle protein increase include feeding (especially extracellular amino acid concentration) and exercise. Paradoxically, during exercise, skeletal muscle protein synthesis is depressed, but it increases again during rest and feeding.

During the catabolic phase of the stress response, muscle wasting occurs as a result of an increase in muscle protein degradation (via enzymatic pathways), coupled with a decrease in muscle protein synthesis. The major site of protein loss is peripheral skeletal muscle, but it also occurs in the respiratory muscles (predisposing the patient to hypoventilation and chest infections) and in the gut (reducing gut motility). Cardiac muscle appears to be mostly spared. The predominant mechanism involved in the wasting of skeletal muscle is the ATP-dependent ubiquitin–proteasome pathway (***Figure 1.4***), although the lysosomal cathepsins and the calcium–calpain pathway play facilitatory and accessory roles.

Under extreme conditions of catabolism (e.g. major sepsis), urinary nitrogen losses can reach 14–20 g/day; this is equivalent to the loss of 500 g of skeletal muscle per day. Muscle catabolism cannot be inhibited fully by providing artificial nutritional support as long as the stress response continues. Hyperalimentation (excess feeding beyond requirements) was once in vogue to try and match the large losses, but it is now recognised that hyperalimentation represents a metabolic stress in itself and that nutritional support should be at a modest level to attenuate rather than replace energy and protein losses. Treating underlying sepsis adequately is fundamental to limiting protein catabolism and is an essential part of effective nutritional support. This includes searching for and treating recurrent septic episodes in the critically ill.

Clinically, a patient with skeletal muscle wasting will experience weakness, fatigue, reduced functional ability, decreased quality of life and an increased risk of morbidity and mortality. In critically ill patients, muscle weakness may be further worsened by the development of critical illness myopathy, a multifactorial condition that is associated with impaired excitation–contraction coupling.

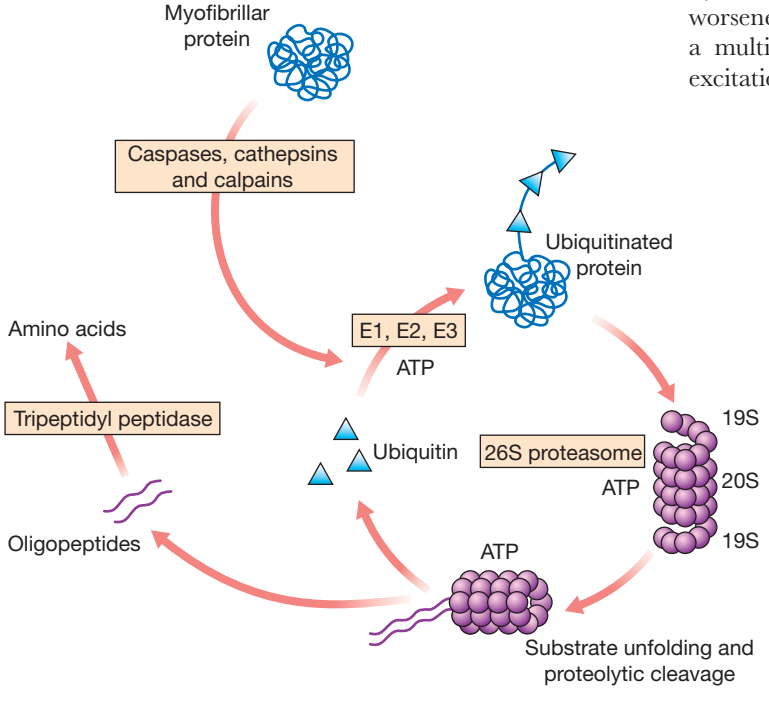

Figure 1.4 The intracellular effector mechanisms involved in degrading myofibrillar protein into free amino acids. The ubiquitin–proteasome pathway is a complex multistep process. ATP, adenosine triphosphate; E1, ubiquitin-activating enzyme; E2, ubiquitin-conjugating enzyme; E3, ubiquitin ligase.

Carl Ferdinand Cori, 1896–1984, and his wife **Gerty Theresa Cori**, 1896–1957, Professors of Biochemistry, Washington University Medical School, St Louis, MI, USA, were awarded a share of the 1947 Nobel Prize for Medicine.

Summary box 1.5

Skeletal muscle wasting

- Provides amino acids for the metabolic support of central organs/tissues
- Is mediated at a molecular level mainly by activation of the ubiquitin–proteasome pathway
- Is inevitable to some degree but is prolonged by sepsis in particular
- Can result in immobility and contribute to prolonged recovery, poor healing, hypostatic pneumonia and death if prolonged and excessive

Alterations in hepatic protein metabolism: the acute-phase protein response

The liver and skeletal muscle together account for >50% of daily body protein turnover. Skeletal muscle has a large mass but a low turnover rate (1–2% per day), whereas the liver has a relatively small mass (1.5 kg) but a much higher protein turnover rate (10–20% per day). Hepatic protein synthesis is divided roughly 50:50 between renewal of structural proteins and synthesis of export proteins. Albumin is the major export protein produced by the liver and is renewed at the rate of about 10% per day. The transcapillary escape rate (TER) of albumin is about 10 times the rate of synthesis, and short-term changes in albumin concentration are most probably due to increased vascular permeability. Albumin TER may be increased threefold following major injury/sepsis. In response to inflammatory conditions, including surgery, trauma and sepsis, proinflammatory cytokines, including IL-1, IL-6 and TNFα and in particular IL-6, promote the hepatic synthesis of positive acute-phase proteins, e.g. fibrinogen and C-reactive protein (CRP). The acute-phase protein response represents a 'double-edged sword' for surgical patients as it provides proteins important for recovery and repair but only at the expense of valuable lean tissue and energy reserves. In contrast to the positive acute-phase reactants, the plasma concentrations of other liver export proteins (the negative acute-phase reactants) fall acutely following injury, e.g. albumin. However, rather than representing a reduced hepatic synthesis rate, the fall in plasma concentration of negative acute-phase reactants is thought principally to reflect increased transcapillary escape, secondary to an increase in microvascular permeability.

Summary box 1.6

Hepatic acute-phase response

The hepatic acute-phase response represents a reprioritisation of body protein metabolism towards the liver and is characterised by:

- **Positive** reactants (e.g. CRP): plasma concentration ↑
- **Negative** reactants (e.g. albumin): plasma concentration ↓

Insulin resistance

Following surgery or trauma, postoperative hyperglycaemia develops as a result of increased glucose production combined with decreased glucose uptake in peripheral tissues. Decreased glucose uptake is a result of insulin resistance, which is temporarily induced within the stressed patient. Suggested mechanisms for this phenomenon include the action of proinflammatory cytokines and the decreased responsiveness of insulin-regulated glucose transporter proteins. The degree of insulin resistance is proportional to the magnitude of the injurious process. Following routine upper abdominal surgery for example, insulin resistance may persist for approximately 2 weeks but this period will extend with prolonged sepsis. Postoperative patients with insulin resistance behave in a similar manner to individuals with type 2 diabetes mellitus. In intensive care, the mainstay of management of insulin resistance is intravenous insulin infusion, which is used to keep blood glucose level within reasonable limits on the basis that this will reduce both morbidity and mortality. However, unduly tight control can increase the risk of significant hypoglycaemia. It should be noted that patients with diabetes whose glycaemic control has been poor prior to their critical illness pose a particular challenge.

CHANGES IN BODY COMPOSITION FOLLOWING INJURY

The average 70 kg male can be considered to consist of fat (13 kg) and fat-free mass (or lean body mass: 57 kg). In such an individual, the lean tissue is composed primarily of protein (12 kg), water (42 kg) and minerals (3 kg) (*Figure 1.5*). The protein mass can be considered as two basic compartments: skeletal muscle (4 kg) and non-skeletal muscle (8 kg), which includes the visceral protein mass. The water mass (42 litres) is divided into intracellular (28 litres) and extracellular (14 litres) spaces. Most of the mineral mass is contained in the bony skeleton.

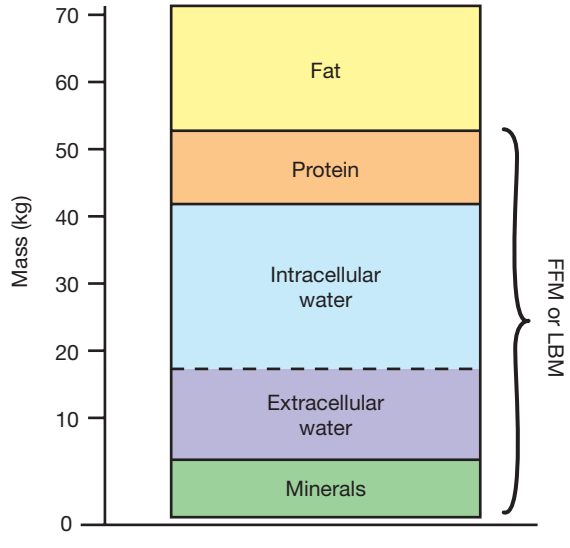

Figure 1.5 The chemical body composition of a normal 70 kg male. FFM, fat-free mass; LBM, lean body mass.

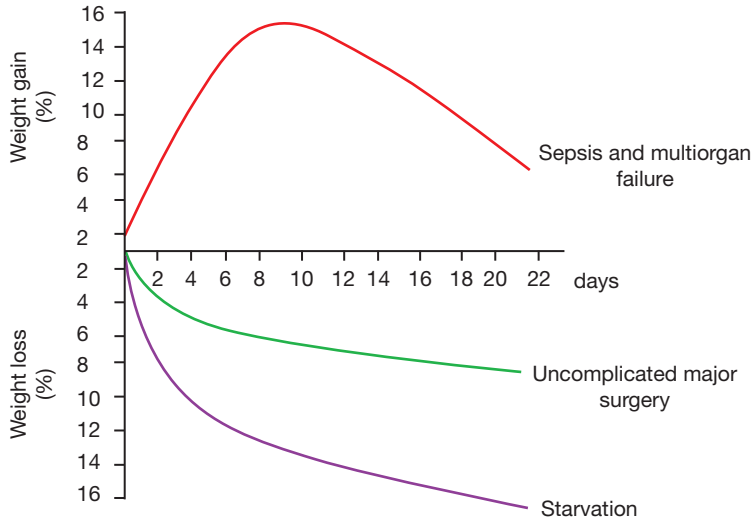

Figure 1.6 Changes in body weight that occur in serious sepsis, after uncomplicated surgery and in total starvation.

The main labile energy reserve in the body is fat, and the main labile protein reserve is skeletal muscle. While fat mass can be reduced without major detriment to function, loss of protein mass results not only in skeletal muscle wasting but also in depletion of visceral protein status. Within lean tissue, each 1 g of nitrogen is contained within 6.25 g of protein, which is contained in approximately 36 g of wet weight tissue. Thus, the loss of 1 g of nitrogen in urine is equivalent to the breakdown of 36 g of wet weight lean tissue. Protein turnover in the whole body is of the order of 150–200 g per day. A normal human ingests about 70–100 g protein per day, which is metabolised and excreted in urine as ammonia and urea (i.e. approximately 14 g N/day). During total starvation, urinary loss of nitrogen is rapidly attenuated by a series of adaptive changes. Loss of body weight follows a similar course (*Figure 1.6*), thus accounting for the survival of hunger strikers for a period of 50–60 days. Following major injury, and particularly in the presence of ongoing septic complications, this adaptive change fails to occur and there is a state of 'auto-cannibalism', resulting in continuing urinary nitrogen losses of 10–20 g N/day (equivalent to 500 g of wet weight lean tissue per day). As with total starvation, once loss of body protein mass has reached 30–40% of the total, survival is unlikely.

Critically ill patients admitted to the intensive care unit with severe sepsis or major blunt trauma undergo massive changes in body composition (*Figure 1.7*). Body weight increases immediately on resuscitation with an expansion of extracellular water by 6–10 litres within 24 hours. Thereafter, even with optimal metabolic care and nutritional support, total body protein will diminish by 15% in the next 10 days, and body weight will reach negative balance as the expansion of the extracellular space resolves. In marked contrast, it is now possible to maintain body weight and nitrogen equilibrium following major elective surgery. This can be achieved by blocking the neuroendocrine stress response with epidural analgesia/other related techniques and providing early oral/enteral feeding. Moreover, the early fluid retention phase can be avoided by careful intraoperative management of fluid balance, with avoidance of excessive administration of intravenous saline.

Summary box 1.7

Changes in body composition following major surgery/critical illness

- Catabolism leads to a decrease in fat mass and skeletal muscle mass
- Body weight may paradoxically increase because of expansion of fluid within the extracellular fluid space

AVOIDABLE FACTORS THAT COMPOUND THE RESPONSE TO INJURY

There are several factors that prolong the acute-phase response to injury (*Table 1.1*) and keep the patient in a catabolic state. Other factors can exacerbate or compound the metabolic stress response both in elective surgery and in the emergency setting. These include anaesthesia, dehydration, starvation (including preoperative fasting), acute medical illness, frailty, chronic diseases or even severe psychological stress (*Figure 1.7*). Attempts to limit or control these factors can also be beneficial to the patient.

Summary box 1.8

Avoidable factors that compound the metabolic response to injury during elective surgery

- Continuing haemorrhage/volume loss
- Hypothermia
- Tissue oedema
- Tissue underperfusion
- Starvation
- Immobility

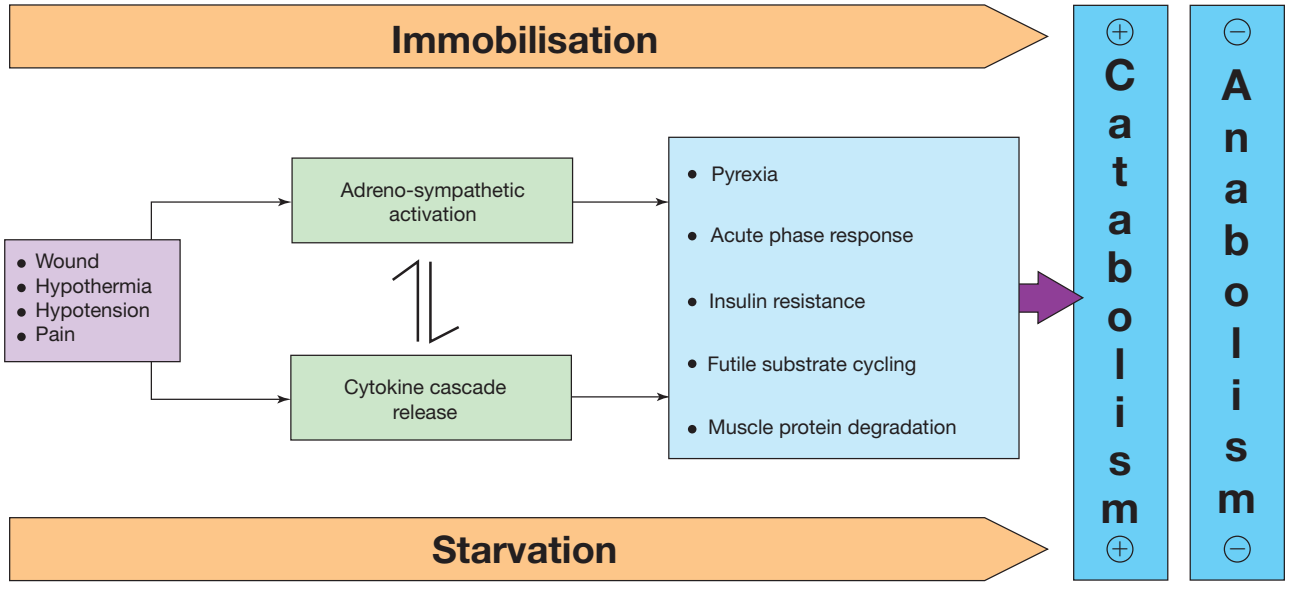

Figure 1.7 Factors that exacerbate the metabolic response to surgical injury include hypothermia, uncontrolled pain, starvation, immobilisation, sepsis and medical complications.

Volume loss

During simple haemorrhage, baroreceptors in the carotid artery and aortic arch and volume receptors in the wall of the left atrium initiate afferent nerve input to the central nervous system, resulting in the release of both aldosterone and antidiuretic hormone (ADH). Pain can also stimulate ADH release. ADH acts directly on the kidney to cause fluid retention. Decreased pulse pressure stimulates the juxtaglomerular apparatus in the kidney and directly activates the renin–angiotensin system, which in turn increases aldosterone release.

Aldosterone causes the renal tubule to reabsorb sodium (and consequently conserve water). ACTH release also augments the aldosterone response. The net effects of ADH and aldosterone result in the natural oliguria observed after surgery and conservation of sodium and water in the extracellular space. The tendency towards water and salt retention is exacerbated by resuscitation with saline-rich fluids. Salt and water retention can result in not only peripheral oedema but also visceral oedema (e.g. in the stomach). Such visceral oedema has been associated with reduced gastric emptying, delayed resumption of food intake and prolonged hospital stay. Careful limitation of intraoperative administration of balanced crystalloids so that there is no net weight gain following elective surgery has been proven to reduce postoperative complications and length of stay.

Hypothermia

Hypothermia results in increased production of adrenal steroids and catecholamines. When compared with normothermic controls, even mild hypothermia results in a two- to threefold increase in postoperative cardiac arrhythmias and increased catabolism. Randomised trials have shown that maintaining normothermia during surgery by an upper body forced-air heating cover reduces wound infections, cardiac complications and bleeding and transfusion requirements.

Tissue oedema

During systemic inflammation, fluid, plasma proteins, leukocytes, macrophages and electrolytes leave the vascular space and accumulate in the tissues as oedema. The oedema can diminish the alveolar diffusion of oxygen and may also impair renal function. Increased capillary leak is mediated by a wide variety of mediators, including cytokines, prostanoids, bradykinin and nitric oxide. Cellular hypoxia and dysfunction can occur. Intracellular volume decreases, and this provides part of the volume necessary to replenish intravascular and extravascular extracellular volume.

Systemic inflammation and tissue underperfusion

The vascular endothelium controls vasomotor tone and microvascular flow and regulates trafficking of nutrients and biologically active molecules. When endothelial activation is excessive, compromised microcirculation and subsequent cellular hypoxia contribute to the risk of organ failure. Controlling the blood sugar appropriately with insulin infusion during critical illness has been proposed to protect the endothelium, probably, in part, via inhibition of excessive iNOS-induced NO release.

Starvation

During starvation, the body is faced with an obligate need to generate glucose to sustain cerebral energy metabolism (100 g of glucose per day). This is achieved in the first 24 hours by mobilising glycogen stores and thereafter by hepatic gluconeogenesis from amino acids, glycerol and lactate. The energy metabolism of other tissues is sustained by mobilising fat from adipose tissue. Such fat mobilisation is mainly dependent on a fall in circulating insulin levels. Eventually, accelerated loss of

lean tissue (the main source of amino acids for hepatic gluconeogenesis) is reduced as a result of the liver converting free fatty acids into ketone bodies, which can serve as a substitute for glucose for cerebral energy metabolism. Provision of 2 litres of intravenous 4% dextrose/0.18% sodium chloride as maintenance intravenous fluids for surgical patients who are fasted provides 80 g of glucose per day and has a significant protein-sparing effect. Avoiding unnecessary fasting in the first instance and early oral/enteral/parenteral nutrition form the platform for avoiding loss of body mass as a result of the varying degrees of starvation observed in surgical patients. Modern guidelines on fasting prior to anaesthesia allow intake of clear fluids up to 2 hours before surgery. Administration of a carbohydrate drink at this time reduces perioperative anxiety and thirst and decreases postoperative insulin resistance.

Immobility

Immobility has long been recognised as a potent stimulus for inducing muscle wasting. Inactivity impairs the normal meal-derived amino acid stimulation of protein synthesis in skeletal muscle. Avoidance of unnecessary bed rest and active early mobilisation are essential measures to avoid muscle wasting as a consequence of immobility. Pre-habilitation programmes provide a better starting point before surgery.

ENHANCED RECOVERY AFTER SURGERY

Modern understanding of the metabolic response to surgical injury and the mediators involved has led to a complete reappraisal of traditional perioperative care and the process known as ERAS. ERAS is evidence based on the strong scientific rationale for avoiding unmodulated exposure to stress, prolonged fasting and excessive administration of intravenous (saline) fluids (***Figure 1.8***). ERAS principles are now applied by protocol to many types of major surgery, bringing considerable benefit in terms of improved outcomes. Reductions in length of hospital stay after surgery of 30–50% are common, with associated savings in healthcare costs. ERAS depends on a multimodal approach where the combined effects of several interventions achieve significant benefits. The widespread adoption of minimal access (e.g. laparoscopic) surgery is a key change in surgical practice that can reduce the magnitude of surgical injury and enhance the rate of patients' return to homeostasis and recovery. Modulating the stress/inflammatory response at the time of surgery may have long-term sequelae over periods of months or longer. For example, β-blockers are associated with improved short- and long-term survival after major surgery, perhaps by modulating the effects of the hyperadrenergic state induced by surgical stress. Equally, in 'open' surgery the use of epidural analgesia to reduce pain, block the cortisol stress response and attenuate postoperative insulin resistance may, via effects on the body's protein economy, favourably affect many of the patient-centred outcomes that are important to postoperative recovery. However, because of the reduction in wound size and tissue trauma, it should be noted that epidural analgesia is no longer recommended for laparoscopic surgery. Patient-controlled analgesia is usually sufficient and avoids the fluid shifts and hypotension seen with epidurals. Adjuncts such as 'one-shot' spinal diamorphine and/or a 6–12-hour infusion of intravenous lidocaine have been suggested to be opiate sparing, to improve gut function and to enhance overall recovery.

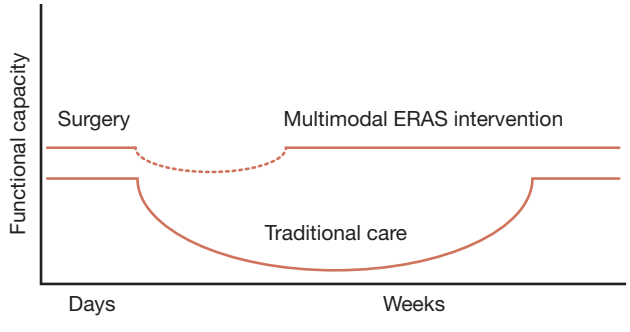

Figure 1.8 Enhanced recovery after surgery (ERAS) programmes use multimodal techniques to limit pain, fluid shifts and tissue damage and to enhance nutrition and rehabilitation in order to minimise the stress response. They have been hugely successful in improving outcomes.

Summary box 1.9

A proactive ERAS approach to prevent unnecessary aspects of the surgical stress response

- Minimal access techniques
- Blockade of afferent painful stimuli (e.g. epidural analgesia, spinal analgesia, wound catheters)
- Minimal periods of starvation
- Early mobilisation

FURTHER READING

Ahl R, Matthiessen P, Sjölin G *et al.* Effects of betablocker therapy on mortality after elective colon cancer surgery: a Swedish nationwide cohort study. *BMJ Open* 2020; **10**: e036164.

Bortolotti P, Faure E, Kipnis E. Inflammasomes in tissue damages and immune disorders after trauma. *Front Immunol* 2018; **9**:1900.

Cole E, Gillespie S, Vulliamy P *et al.* Multiple organ dysfunction after trauma. *Br J Surg* 2020; **107**: 402–12.

Fearon KCH, Ljungqvist O, von Meyenfeldt M *et al.* Enhanced recovery after surgery: a consensus review of clinical care for patients undergoing colonic resection. *Clin Nutr* 2005; **24**: 466–77.

Huber-Lang M, Lambris JD, Ward PA. Innate immune responses to trauma. *Nat Immunol* 2018; **19**(4): 327–41.

Ljungqvist O. Insulin resistance and outcomes in surgery. *J Clin Endocrinol Metab* 2010; **95**: 4217–19.

Ljungqvist O, Scott M, Fearon KCH. Enhanced recovery after surgery: a review. *JAMA Surg.* 2017; **152**(3): 292–8.

Mira J, Cuschieri J, Ozrazgat-Baslanti T *et al.* The epidemiology of chronic critical illness after severe traumatic injury at two level-one trauma centers. *Crit Care Med* 2017; **45**(12): 1989–96.

Vanhorebeek O, Langounche L, Van den Berghe G. Endocrine aspects of acute and prolonged critical illness. *Nat Clin Pract Endocrinol Metab* 2006; **2**: 20–31.

Vourc'h M, Roquilly A, Asehnoune K. Trauma-induced damage-associated molecular patterns-mediated remote organ injury and immunosuppression in the acutely ill patient. *Front Immunol* 2018; **9**: 1330.

Wilmore DW. From Cuthbertson to fast-track surgery: 70 years of progress in reducing stress in surgical patients. *Ann Surg* 2002; **236**: 643–8.